

Topic 4.3

4.3: Destructors

A destructor is nearly identical in form to a constructor:

```
class Distance
{
    ...
    Distance(void);    // Constructor.
    ~Distance(void);   // Destructor.
};
```

The tilde ~ preceding the class name indicates this is the *destructor* for a class.

The destructor for a class is invoked automatically by the compiler whenever an object of that class ceases to exist, that is, its extent is terminated.

For ordinary objects, this is not too useful. However, class objects can contain dynamically allocated memory, and destructors are a useful place to reclaim this space.

In the following example, assume a destructor exists for class [Distance](#):

```
void qrs( )
{
    Distance auto_d;
    Distance *dynamic_d;

    dynamic_d = new Distance;
    ...

    // Prior to returning, the destructor
    // is called for auto_d.
    // The destructor is not called for dynamic_d
    // because it is truly dynamically-allocated
    // and the compiler does not know when its
    // extent terminates.
}
```

Topic 4.4 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-4>)